ISO 29148 • EARS • IEEE

PIPELINE 1 — Surgical Requirements Quality Analyzer (5 Staged Layers)

Input Contract: Requirement(id, type, section, text) + SRSContext(domain, glossary, window±3, standards)

Audits individual requirement in context without leaking future specification data.

LAYER 1 — Document Scorer (ISO 29148 7-Attribute Continuous Scoring)

• Unambiguity, Completeness, Consistency, Conformance, Correctness, Singularity, Verifiability

Deterministic $Q_{req} = \Sigma\, w_i \cdot S(A_i) \in [0,10]$ | Shortcoming emitted if $S(A_i) < 8.0$

Emits D_L1 candidate defects (Zero filtering at this stage)

LAYER 2 — Defect Reasoner (Contextual False-Positive Suppression)

• Resolves the Contextual False-Positive Paradox via glossary, boundary & neighbor cross-check.

✕ Discarded: Resolved in surrounding context (FSR = |D_discarded| / |D_L1|) ➔ Terminated

✓ Retained: Genuine defects (D_retained) ➔ ONLY these proceed downstream

LAYER 3 — Solution Generator (EARS-Compliant Surgical Refinement)

• Synthesizes minimal surgical edits: preserves system names, domain entities & business intent.

• Applies EARS patterns (Ubiquitous, Event, State) to address ONLY retained defect IDs.

Emits refined text, itemized changes, and intent preservation rationale.

LAYER 4 — Solution Validator (Fail-Closed Adversarial Critic)

• 5 Fail-Closed Checks: Defect Resolution • No Ambiguities • No Unwarranted Assumptions

• Functional Intent Preservation • Syntactic Conformance

Conjunctive Gating: $V = \bigwedge c_j$ ➔ Fail-closed rejection if any check fails

LAYER 5 — Decision Manager (Word Diff & Review Governance)

• Generates punctuation-preserving word diff highlighting exact surgical replacements.

• Packages full chain-of-custody audit record for stakeholder sign-off.

Emits PipelineExecutionResult(status="completed"|"error", proposal, time)

Pipeline 1 Resulting Proposal:

$R_{prop} = \{\, u \in U \mid \text{Refined}(u) \neq \text{Original}(u) \wedge \text{Status} = \text{"completed"} \,\}$

INDEPENDENT PROPOSALS

PIPELINE 2 — Empirical Evolution Benchmark vs Final SRS Release

Document Ground Truth Inputs:

• **Draft SRS (V_N):** Early specification evaluated blindly by Pipeline 1

• Final Version of SRS (V_final / V_N+1): Official release ground truth

1. Structural Document Aligner (aligner.py)

• Two-Stage Alignment: Stage 1 Exact ID + Stage 2 Normalized Levenshtein

• Synthetic ID Isolation: Prevents unnumbered items collapsing into "REQ"

2. Evolutionary Change Classifier (change_classifier.py)

• 6 Exclusive Classes: Unchanged • Editorial • Scope Change • Added • Deleted

• Quality Refinement: Ambiguity, Parameterization, Scope Clarification, Consistency

3. Semantic Correspondence Evaluator (evaluator.py)

Chain-of-thought qualitative audit comparing Pipeline-1 proposal vs Final SRS text

MATCHED: Independently anticipated semantically equivalent fix in Final SRS.

PARTIALLY_MATCHED: Targeted correct defect clause; alternative resolution.

NOT_MATCHED: Uncorrelated change, extraneous edit, or pipeline crash.

Fast Short-Circuits: Pipeline error ➔ NOT_MATCHED | Both unchanged ➔ MATCHED

4. Bounded Benchmark Evaluation Metrics:

Strict Match Rate (PMR_strict):
$|R_{prop} \cap R_{MATCHED}| / |R_{prop}|$

Combined Match (PMR_comb):
$|R_{prop} \cap (R_M \cup R_{PART})| / |R_{prop}|$

Quality Refinement Recall (QRR):
$|R_{prop} \cap H_{qual} \cap R_M| / |H_{qual}|$

Quality Precision (QRP):
$|R_{prop} \cap H_{qual} \cap R_M| / |R_{prop}|$

Overall Evolutionary Agreement (OEA):
$(|R_{prop} \cap R_{MATCHED}| + |U_{TN}|) / |U|$ [True Negatives strictly separated]